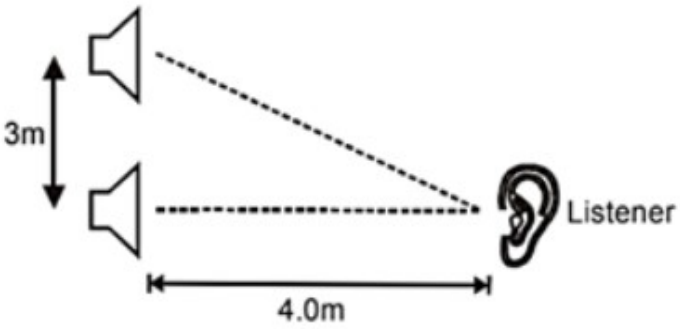
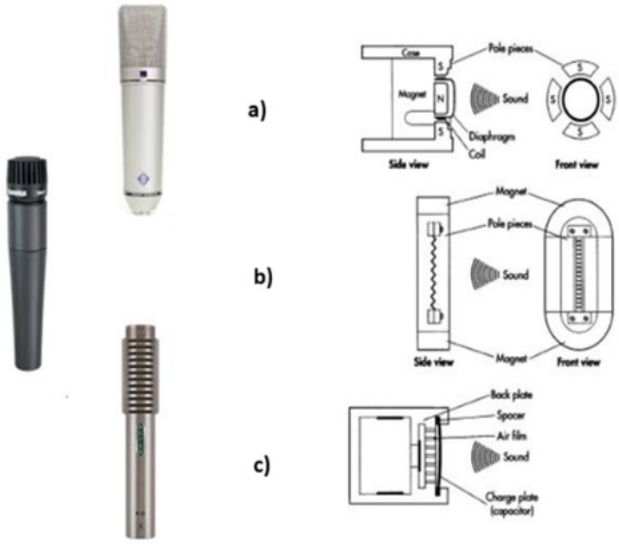
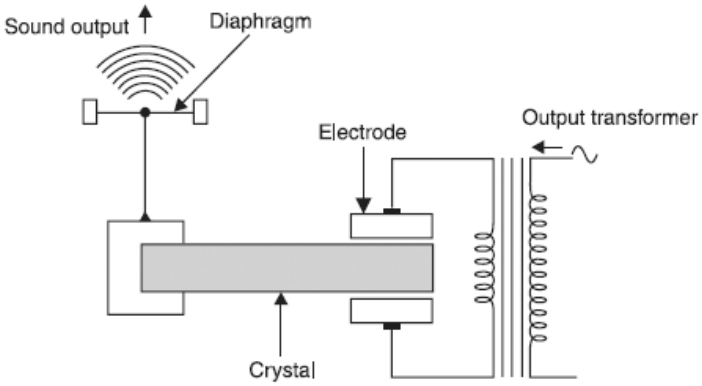
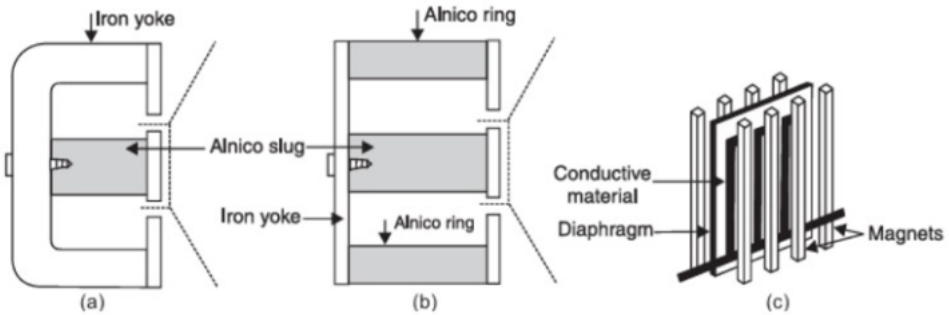
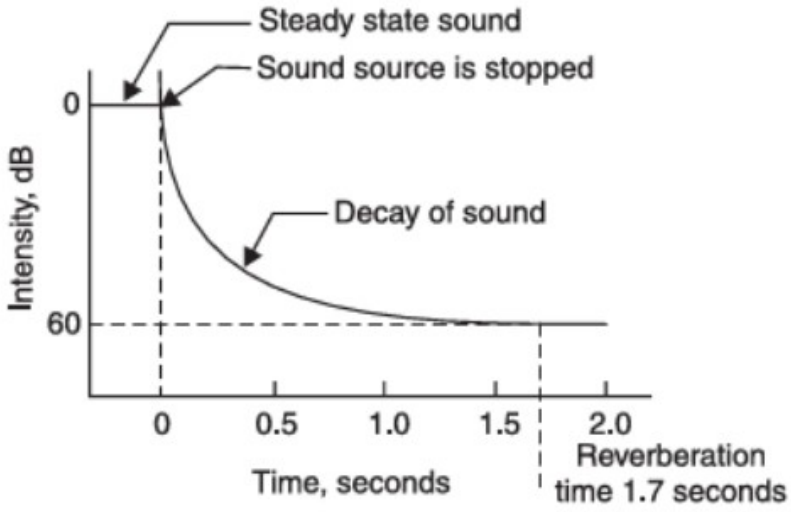
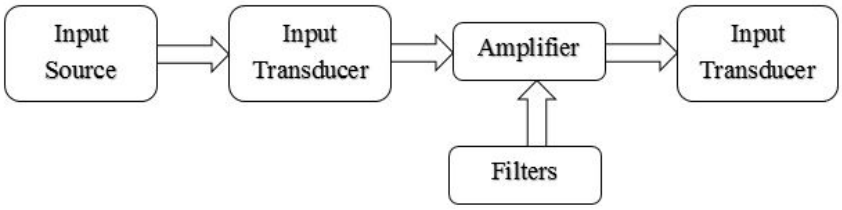
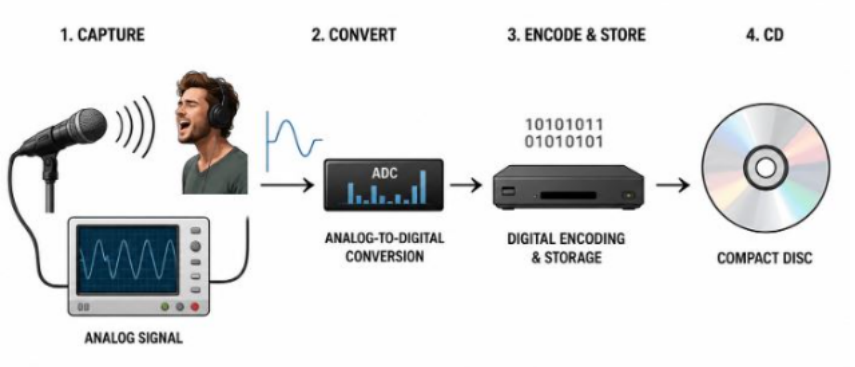
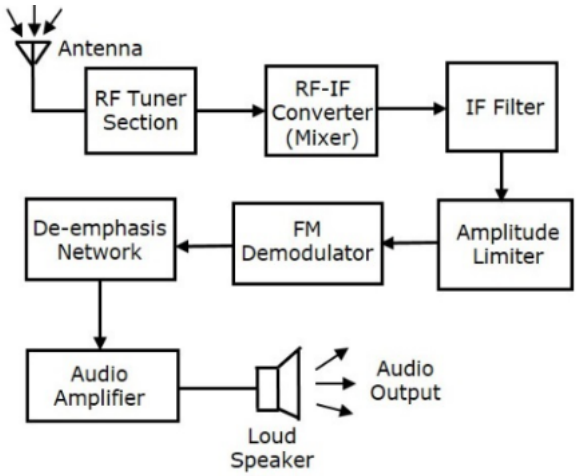
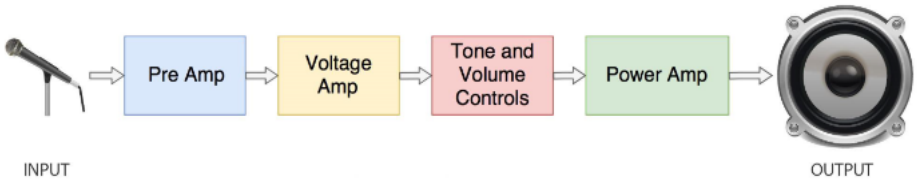
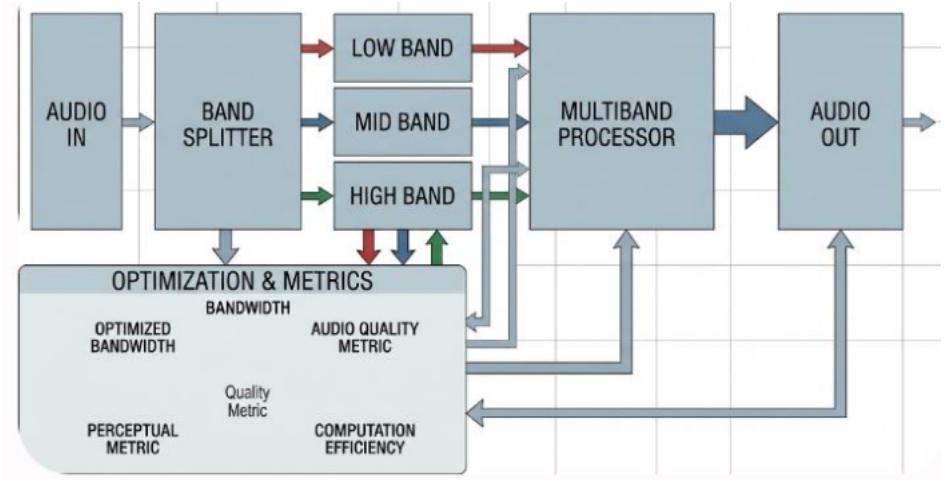
Q. No.	Questions
UNIT 1	**SECTION 1**
1	Two loudspeakers connected to the same audio source are arranged in an auditorium as shown in the figure.  Figure: Two loudspeakers connected to the same audio source Identify the regions of constructive and destructive interference and explain how they influence sound intensity. (5 Marks)
2	A professional recording studio is evaluating dynamic, condenser, and ribbon microphones to achieve accurate sound capture for different recording requirements. The internal structures of dynamic (moving-coil), condenser, and ribbon microphones are shown in the figure.  Figure: Internal structures of these microphones Identify the key components responsible for sound transduction in each microphone and illustrate how the sound-to-electrical signal conversion process differs among them. (5 Marks)
3	An audiologist measures the hearing thresholds of a patient's right ear as shown below. <table><tr><td>Frequency (Hz)</td><td>500</td><td>1000</td><td>2000</td><td>4000</td></tr><tr><td>Air Condition (dB HL)</td><td>45</td><td>50</td><td>55</td><td>60</td></tr><tr><td>Bone Condition (dB HL)</td><td>10</td><td>10</td><td>15</td><td>15</td></tr></table> Analyze the hearing test results to identify the air-bone gap at each frequency, the average air-bone gap,

	and hearing loss. (5 Marks)	
4	(i) A newly installed recording system produces weak audio and distorted sound even though all components are functioning correctly. They are identifies that the problem is related to the electrical compatibility of the connected devices. Infer the reason for the poor system performance and justify a suitable solution. (3 Marks)	
	(ii) Two people seated opposite each other are being interviewed using a single microphone. The recording should capture both voices equally while minimizing sounds from the sides. Justify the microphone directivity suitable for this recording. (2 Marks)	
UNIT 1	**SECTION 2**	
5	A hobbyist uses a crystal loudspeaker with a capacitance of 0.05 μ F as a full-range speaker (100 Hz-10,000 Hz). The crystal loudspeaker is shown in the figure.  Figure: Crystal Loudspeaker The capacitive reactance is given by Xc = 1/2 π fc A crystal loudspeaker is evaluated at 100 Hz and 10,000 Hz. Apply the reactance formula and infer its suitability as a full-range driver. (5 Marks)	
6	A design team is developing a three-way hi-fi speaker system consisting of a woofer, mid-range, and tweeter. Two diaphragm sizes (10-inch and 3-inch) and two magnet strengths (moderate and maximum) are available.  Figure. Shapes of magnets in PM speakers a) Predict the suitable diaphragm size and magnet strength for the woofer, mid-range, and tweeter based on the design requirements. b) Compare the performance of a woofer using maximum magnetic flux with one using moderate magnetic flux.	

	(5 Marks)
7	An acoustics consultant designs a lecture hall with a volume of 300 m^3. The hall has the following materials: Acoustic plaster boards: 150 m^2, absorption coefficient = 0.25 Wooden flooring: 80 m^2, absorption coefficient = 0.09 Audience seating: 60 m^2, absorption coefficient = 0.46  Figure: Growth and decay of sound in a room Referring to Figure, Use Sabine's equation to compute the total absorption and reverberation time, and assess the acoustic quality of the hall. (5 Marks)
8	(i) A sound system installer is comparing two candidate speakers for a paging system, both driven from the same amplifier output of 40 watts of electrical power. Speaker X is a large, high-efficiency type rated at 25 percent efficiency. Speaker Y is a small bookshelf-type speaker rated at only 2 percent efficiency. Apply the given efficiency percentages to identify the acoustic sound power output each speaker would produce from the same 40-watt electrical input, and which speaker produces more usable acoustic power. (3 Marks)
	(ii) Justify whether Speaker X should be selected only because it produces higher acoustic power. (2 Marks)
UNIT 2	**SECTION 1**
9	A hospital announcement system experiences a low-frequency hum from nearby generators, reducing voice clarity. Audio filters are used to attenuate unwanted frequencies and improve audio quality, as shown in the figure.  Figure : Block diagram of an audio system Outline how audio filters improve signal quality. Compare High Pass, Low Pass, Band Pass, and Band Stop filters, and recommend the most suitable filter(s) for reducing low-frequency noise and maintaining clear speech. (5 Marks)

10	A CD recording system converts an analog microphone signal into digital data through sampling, quantization, CIRC error correction, EFM, synchronization, and optical recording using laser-generated pits and lands, as shown in the figure.  Figure : The analog to digital conversion using CD path Infer the purpose of each stage PCM, quantization, CIRC, EFM and Laser recording in the block diagram of the audio processing sequence from analog input to CD storage . (5 Marks)
11	An FM broadcasting station receiver shown in the figure below experiences excessive background noise at higher audio frequencies during transmission.  Figure : FM receiver Three options are given below Option A: Increase transmitter power. Option B: Use pre-emphasis at the transmitter and matching de-emphasis at the receiver. Option C: Increase receiver amplifier gain. Select and justify the best option based on its ability to improve SNR while preserving audio quality. (5 Marks)
12	(i) A listener reports the following observations: • Sound appears to originate from both the front and rear. • Audience ambience is clearly audible behind the listener. • Four independent speakers are used. Working backwards from these observations, identify the audio reproduction system being used and infer how the recording and playback arrangement produces this listening experience. (3 Marks)
	(ii) Using an everyday real-life analogy that watching a football match with one eye provides limited depth perception, whereas using both eyes allows accurate judgment of position and distance., Indicate why

	stereophonic sound creates a stronger sense of direction than monophonic sound. (2 Marks)
UNIT 2	**SECTION 2**
13	A recording studio upgrades its audio console. Speech is clear, but music lacks bass depth and high-frequency brilliance. The system includes an integrated stereo amplifier with signal-conditioning and amplification stages, as shown in the figure.  Figure : Amplifier circuit Without replacing the loudspeakers: a)Infer the circuit modification required to enhance low- and high-frequency components. b)Indicate how the audio signal is processed and amplified before reaching the loudspeakers. c)Outline the suitable signal-conditioning approach based on frequency control and amplification. (5 Marks)
14	A digital audio system reduces data by dividing the input signal into frequency bands, processing each band independently, and recombining them at the receiver.  Figure : Consolidated audio processing syetm a) Illustrate the functional block diagram from input to reconstructed output. b) Infer the need for each major processing stage. (5 Marks)
15	A student prepares the following simplified encoder diagram shown in the figure below.

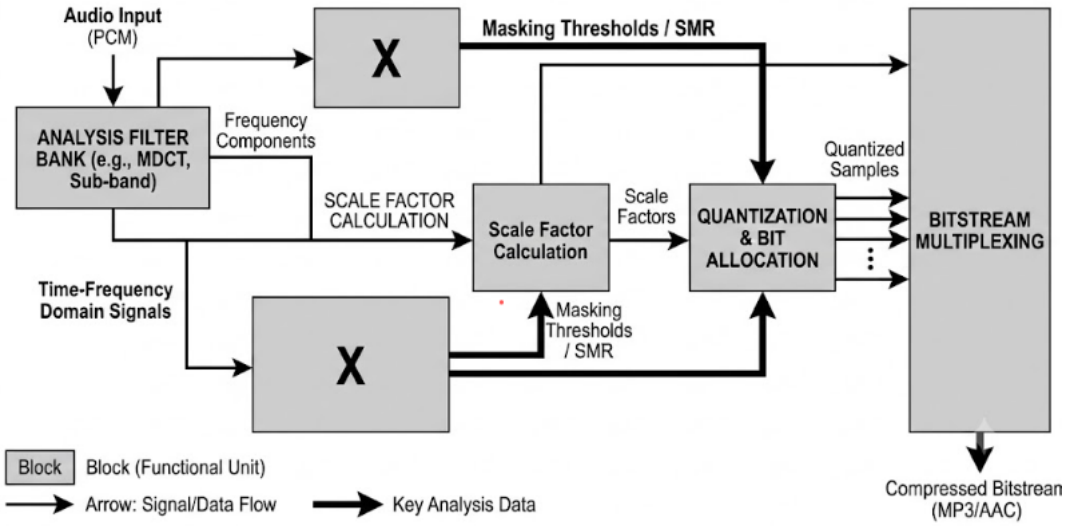


Figure: simplified encoder diagram

a) Interpret the effect of the absence of the sampling and quantization block on the encoder operation.

b) Indicate the way by which sampling and quantization block improves coding efficiency.

c) Infer how bit allocation is distributed among multiple data streams in a bit stream multiplexing system to achieve efficient transmission.

(5 Marks)

16

(i) A music system produces a very weak audio signal from a microphone before sending it to the loudspeaker. To obtain clear and loud sound, the signal passes through different stages of an integrated stereo amplifier.

Outline how the preamplification, tone control, and power amplification stages process the low-level audio signal.

(3 Marks)

(ii) An integrated stereo amplifier includes several functional stages to process audio before output. Infer which specific circuit is responsible for adjusting the relative signal amplitudes between the left and right audio channels, and justify your answer.

a) Preamplifier

b) Tone Control Circuit

c) Balance Control

d) Loudspeaker

(2 Marks)

UNIT 3 | **SECTION 1**

17

The description of a television system is as follows:

The camera directly displays the optical image on the television screen, as shown in the figure below. Since the human eye can see continuous motion, scanning is unnecessary. The transmitter sends the entire image as one signal, and synchronization is not important because the receiver can reconstruct the picture automatically.

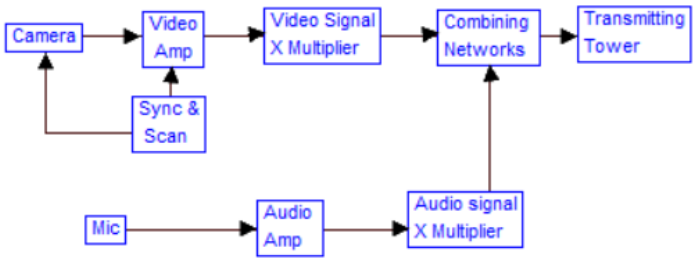


Figure : simple TV system

Identify the conceptual errors and rewrite the explanation to correctly refer to the roles of image conversion, scanning, transmission, synchronization, and image reconstruction.

(5 Marks)

18 A television camera captures a scene with varying light and dark regions in space and time, while the transmission channel carries a time-varying signal. The scanning beam moves across the image is shown in the figure below, converting brightness variations into an electrical signal. At the receiver, this signal is used to reconstruct the image.

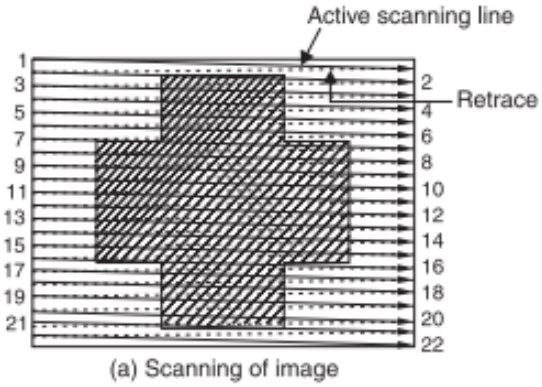


Figure : Scanning of an image

Analyze why scanning is essential in television systems, how it enables picture information to be transmitted through a single channel, and how scanning coverage affects the detail of the reproduced image.

(5 Marks)

19 To achieve good picture quality, a television system must balance scanning lines, frame repetition rate, flicker, and bandwidth. The waveform of individual scanning lines and an image with 32 line scans is shown in the figure below.

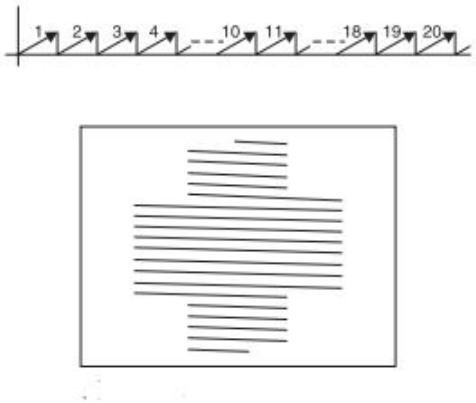


Figure : Waveform of individual scanning lines and image with 32 line scan

a) Infer how increasing the number of scanning lines affects picture resolution.

	b) Interpret how persistence of vision makes moving images appear continuous. c) Indicate how persistence of vision makes moving images appear continuous. (5 Marks)
20	(i) It is believed that sitting extremely close to the screen will always allow more picture details to be resolved because the picture appears physically larger. From a visual perspective, the minimum angle of resolution and the comfortable viewing distance indicate why this conclusion is not necessarily correct. (3 Marks)
	(ii) The observation of differences in picture clarity depending on the scanning process, visual resolution, and viewing position. Identify the two parameters that influence vertical picture detail and the eye's ability to distinguish fine details. (2 Marks)